SYSTEM ENGINEER | ONLINE ASSESSMENT

Infosys
Numerical Puzzle

Top 30 Practice Questions with Answers

Series, grids, cryptarithmetic, Sudoku, logic and counting puzzles.

30	4	100%
Questions	Options each	With solutions

WHAT'S INSIDE

Number Series
9 questions
Spot the rule and find the next term

Grids & Magic Squares
5 questions
Row and column rules, magic sums

Sudoku & Cryptarithm
3 questions
Digit logic and letter-to-number puzzles

Pyramids & Operators
2 questions
Brick sums and symbol substitution

Weighing, Clock, Calendar
4 questions
Classic reasoning with numbers

Digits & Counting
7 questions
Number properties and combinations

HOW TO USE

Give each question about 2 minutes, then check the answer and the solution below it.

Infosys Numerical Puzzle

Top 30 Practice Questions

1 In the cryptarithm SEND + MORE = MONEY, each letter stands for a different digit. Find the value of M + O + N + E + Y.

- A) 12
- B) 14
- C) 16
- D) 18

Answer: B) 14

9567 + 1085 = 10652. So M = 1, O = 0, N = 6, E = 5 and Y = 2, and their sum is 14.

2 Find the next term in the series: 3, 8, 18, 38, 78, ?

- A) 146
- B) 158
- C) 152
- D) 162

Answer: B) 158

Each term is double the previous term plus 2. So 78 × 2 + 2 = 158.

3 In this 3 × 3 magic square every row, column and diagonal has the same sum. Find the sum of the three missing numbers.

8	?	6
3	5	?
?	9	2

- A) 10
- B) 14
- C) 15
- D) 12

Answer: D) 12

The magic sum is 15 (middle row 3 + 5 + 7). The missing numbers are 1, 7 and 4, so 1 + 7 + 4 = 12.

4 Every row follows the same rule. Find the missing number.

3	4	25
5	12	169
8	15	?

A) 221

B) 289

C) 255

D) 306

Answer: B) 289

In each row, (first)² + (second)² = third. So 8² + 15² = 64 + 225 = 289.

5 Find the missing term: 2, 6, 12, 20, 30, 42, ?

A) 52

B) 54

C) 58

D) 56

Answer: D) 56

The differences are 4, 6, 8, 10, 12, so the next difference is 14. Hence 42 + 14 = 56.

6 Among 8 identical-looking coins, one is slightly heavier. Using a balance scale, what is the minimum number of weighings that guarantees finding it?

A) 2

B) 1

C) 3

D) 4

Answer: A) 2

Split the coins into 3, 3 and 2. Weigh 3 against 3 to find the heavy group, then one more weighing finds the coin. Two weighings are enough.

7 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box contains the digits 1 to 4. Find the sum of the four missing digits.

1	2	?	4
?	4	1	2
2	?	4	3
4	3	2	?

A) 6

B) 8

C) 7

D) 9

Answer: B) 8

The completed rows are 1 2 3 4, 3 4 1 2, 2 1 4 3 and 4 3 2 1. The missing digits are 3, 3, 1 and 1, so the sum is 8.

8 Find the next term in the series: 2, 3, 5, 9, 17, 33, ?

A) 65

B) 63

C) 64

D) 67

Answer: A) 65

Each term is double the previous term minus 1. So 33 × 2 - 1 = 65.

9 What is the smaller angle between the hour hand and the minute hand of a clock at 3:40?

A) 110

B) 130

C) 120

D) 140

Answer: B) 130

At 40 minutes the minute hand is at 240°. The hour hand is at 90° + 20° = 110°. The angle is 240° - 110° = 130°.

10 Every row follows the same rule. Find the missing number.

2	3	12
4	5	40
6	7	?

A) 84

B) 42

C) 78

D) 96

Answer: A) 84

In each row, third number = first × second × 2. So 6 × 7 × 2 = 84.

11 In a number pyramid, each brick is the sum of the two bricks directly below it. If the bottom row is 5, 9, 4, 7, what number is at the top?

A) 48

B) 49

C) 53

D) 51

Answer: D) 51

Second row: 14, 13, 11. Third row: 27, 24. Top brick: 27 + 24 = 51.

12 Find the next term in the series: 7, 10, 8, 11, 9, 12, ?

A) 9

B) 11

C) 10

D) 13

Answer: C) 10

The pattern is +3, -2, +3, -2, and so on. After 12 comes 12 - 2 = 10.

Infosys Numerical Puzzle

13 This 3 × 3 grid is a magic square (all rows, columns and diagonals have the same sum). Find the number in the bottom-middle cell.

9	2	?
?	6	8
5	?	3

A) 8

B) 9

C) 10

D) 12

Answer: C) 10

The completed square is 9 2 7 / 4 6 8 / 5 10 3 with a magic sum of 18. Bottom row: 18 - 5 - 3 = 10.

14 Find the next term in the series: 5, 11, 23, 47, 95, ?

A) 189

B) 190

C) 191

D) 193

Answer: C) 191

Each term is double the previous term plus 1. So 95 × 2 + 1 = 191.

15 If 1st January 2026 is a Thursday, which day of the week will 1st January 2028 be?

A) Friday

B) Saturday

C) Sunday

D) Monday

Answer: B) Saturday

2026 and 2027 are not leap years, so the gap is 730 days = 104 weeks + 2 days. Thursday + 2 days = Saturday.

16 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box contains the digits 1 to 4. Find the sum of the six missing digits.

2	?	4	3
?	3	?	1
1	2	?	4
3	?	1	?

A) 16

B) 14

C) 15

D) 17

Answer: A) 16

The completed rows are 2 1 4 3, 4 3 2 1, 1 2 3 4 and 3 4 1 2. The missing digits are 1, 4, 2, 3, 4 and 2, so the sum is 16.

17 Find the next term in the series: 4, 9, 25, 49, 121, 169, ?

A) 225

B) 256

C) 324

D) 289

Answer: D) 289

These are the squares of prime numbers: 2^2, 3^2, 5^2, 7^2, 11^2, 13^2. The next prime is 17, so $17^2 = 289$.

18 How many three-digit numbers have a digit sum equal to 4?

A) 8

B) 12

C) 10

D) 15

Answer: C) 10

The numbers are 103, 112, 121, 130, 202, 211, 220, 301, 310 and 400, which gives 10 numbers.

19 Every column follows the same rule. Find the missing number.

5	8	6
3	4	2
15	32	?

A) 12

B) 10

C) 14

D) 18

Answer: A) 12

In each column, top number × middle number = bottom number. So 6 × 2 = 12.

20 Find the next term in the series: 3, 6, 7, 14, 15, 30, ?

A) 28

B) 31

C) 32

D) 33

Answer: B) 31

The pattern alternates between × 2 and + 1. After 30 comes 30 + 1 = 31.

21 If × means +, + means −, and − means ×, what is the value of 12 × 8 + 6 − 2?

A) 4

B) 12

C) 8

D) 20

Answer: C) 8

Replacing the signs gives 12 + 8 − 6 × 2 = 12 + 8 − 12 = 8.

22 Find the next term in the series: 2, 12, 36, 80, 150, ?

- **A** 240
- **B** 256
- **C** 264
- **D** 252

Answer: D) 252

The terms follow $n^2 \times (n + 1)$. For n = 6: 36 × 7 = 252.

23 Among 12 coins, one is either heavier or lighter than the others (you do not know which). What is the minimum number of weighings on a balance scale needed to find the odd coin and know whether it is heavy or light?

- **A** 2
- **B** 4
- **C** 5
- **D** 3

Answer: D) 3

This is the classic 12-coin problem. Three weighings give 27 outcomes, enough to cover 24 cases (12 coins × heavy or light).

24 At a meeting, 12 people each shake hands exactly once with every other person. How many handshakes take place in total?

- **A** 60
- **B** 66
- **C** 72
- **D** 132

Answer: B) 66

Number of handshakes = 12 × 11 ÷ 2 = 66.

25 Find the next term in the series: 1, 2, 6, 24, 120, ?

A 600

B 700

C 720

D 840

Answer: C) 720

These are factorials: 1!, 2!, 3!, 4!, 5!. The next term is 6! = 720.

26 How many two-digit numbers are equal to 4 times the sum of their digits?

A 3

B 4

C 5

D 6

Answer: B) 4

10a + b = 4(a + b) gives 6a = 3b, so b = 2a. The numbers are 12, 24, 36 and 48, which is 4 numbers.

27 What is the smallest three-digit number that is divisible by 4, 6 and 9?

A 108

B 100

C 126

D 144

Answer: A) 108

The LCM of 4, 6 and 9 is 36. Its multiples are 36, 72, 108, so the smallest three-digit one is 108.

28 What is the last digit of 7 raised to the power 2026?

- A) 9
- B) 1
- C) 3
- D) 7

Answer: A) 9

The last digits of powers of 7 repeat as 7, 9, 3, 1. Since 2026 leaves remainder 2 when divided by 4, the last digit is 9.

29 If letters take their alphabet positions (A = 1, B = 2, ... Z = 26), what is the total value of the word DOG?

- A) 26
- B) 24
- C) 25
- D) 27

Answer: A) 26

D = 4, O = 15 and G = 7. So the total is 4 + 15 + 7 = 26.

30 A two-digit number has a digit sum of 9. When its digits are reversed, the number increases by 27. Find the number.

- A) 27
- B) 45
- C) 36
- D) 54

Answer: C) 36

Let the number be 10a + b. Then a + b = 9 and 9(b - a) = 27, so b - a = 3. Hence a = 3 and b = 6, giving 36.

Quick Answer Key

Q1 B — 14	**Q2** B — 158	**Q3** D — 12	**Q4** B — 289	**Q5** D — 56
Q6 A — 2	**Q7** B — 8	**Q8** A — 65	**Q9** B — 130	**Q10** A — 84
Q11 D — 51	**Q12** C — 10	**Q13** C — 10	**Q14** C — 191	**Q15** B — Saturday
Q16 A — 16	**Q17** D — 289	**Q18** C — 10	**Q19** A — 12	**Q20** B — 31
Q21 C — 8	**Q22** D — 252	**Q23** D — 3	**Q24** B — 66	**Q25** C — 720
Q26 B — 4	**Q27** A — 108	**Q28** A — 9	**Q29** A — 26	**Q30** C — 36